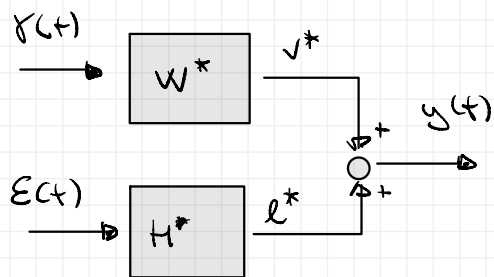


Exercice 5



$$x(t) \sim \text{WN}(1, 1)$$

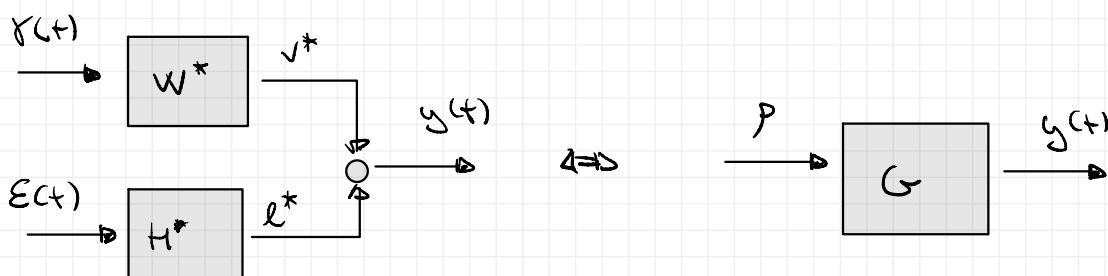
$$x \perp e$$

$$e(t) \sim \text{WN}(-1, 1)$$

$$W^* = \frac{1}{1 - 1/4 z^{-2}}$$

$$H^* = \frac{-2z^{-1}}{1 - 1/4 z^{-1}}$$

⇒ Find, if possible, an equivalent SWSL-NOISE STOCHASTIC DESCRIPTION OF $y(t)$



TO FIND AN EQUIVALENT PROCESS WE NEED TO BE SURE THAT WITH THIS TWO DESCRIPTIONS $y(t)$ HAS:

1. SAME MEAN VALUE
2. SAME COVARIANCE FUNCTION / PSD

LET'S START ADDRESSING (2), BY COMPUTING $G(z)$, λ_p^2 SUCH THAT THIS PSD ARE IDENTICAL

$$\Gamma_{y,2}(\omega) = \Gamma_{y,1}(\omega)$$

$$\Gamma_{y,2}(\omega)$$

IN PREVIOUS EXERCISES WE SAW THAT

IF THE NOISES ARE ORTHOGONAL

$$y_y(z) = y_{v^*}(z) + y_{l^*}(z) + \text{O}$$

↓

$$\Gamma_{y,2}(\omega) = \Gamma_{v^*}(\omega) + \Gamma_{l^*}(\omega)$$

$$\begin{aligned} P_{y^*}(\omega) &= |W^*(e^{j\omega})|^2 \cdot P_\gamma \\ &= \frac{1}{17/16 - 1/2 \cos \omega} \end{aligned}$$

$$\begin{aligned} P_{e^*}(\omega) &= |H^*(e^{j\omega})|^2 \cdot P_\varepsilon \\ &= \frac{4}{17/16 - 1/2 \cos \omega} \end{aligned}$$

$\Downarrow$

$$P_{y,2}(\omega) = \frac{5}{17/16 - 1/2 \cos \omega}$$

LET'S ADDRESS $P_{y,1}(\omega)$:

$$P_{y,1} = |G(e^{j\omega})|^2 \cdot P_p$$

FOR A GENERAL $G(z)$, HAND COMPUTATIONS ARE HARD.

FOR "SIMPLE" SYSTEMS, FOR INSTANCES ARMA(1,1) COMPUTATIONS ARE NOT SO DIFFICULT

$$\text{ARMA}(1,1): y(t) = \underbrace{\frac{1 + c z^{-1}}{1 + a z^{-1}}}_{\hookrightarrow G(z)} \cdot p(t)$$

$$\begin{aligned} P_{y,1}(\omega) &= \frac{1 + c e^{-j\omega}}{1 + a e^{-j\omega}} \cdot \frac{1 + c e^{j\omega}}{1 + a e^{j\omega}} \cdot \lambda_p^2 \\ &= \frac{(1+c^2) + 2c \cdot \cos \omega}{(1+a^2) + 2a \cdot \cos \omega} \cdot \lambda_p^2 \end{aligned}$$

~ PSD FOR A GENERIC ARMA(1,1) PROCESS.

ITS STRUCTURE FITS WITH THE PSD WE WANT TO MATCH!

SO WE CAN ENFORCE THE EQUALITY AND FIND THE PARAMETERS OF $G(z)$, $p(t)$

$$\frac{5}{17/16 - 1/2 \cos \omega} = \frac{(1+c^2) + 2c \cdot \cos \omega}{(1+a^2) + 2a \cdot \cos \omega} \cdot \lambda_p^2$$

$$\bullet c = 0$$

$$\bullet \lambda_p^2 = 5$$

$$\bullet \left. \begin{aligned} 1+a^2 &= 17/16 \\ 2a &= -1/2 \end{aligned} \right\}$$

$$\Rightarrow a = -1/4$$

$$G(z) = \frac{1}{1 - 1/4 z^{-1}}$$

$$\lambda_p^2 = 5$$

LST'S ADDRESS 1. To do that, we impose the equivalence of the expected values of $y(t)$

$$\mu_{y,2} = \mu_{y,1}$$

$$W^*(z=1) \cdot \mu_y + H^*(z=1) \cdot \mu_E = G(z=1) \cdot \mu_p$$

$$\frac{4}{3} \cdot 1 + \frac{4}{3} \cdot 2 = G(z=1) \cdot \mu_p$$

$$4 = \frac{4}{3} \cdot \mu_p$$

$\Downarrow$

$$\mu_p = \frac{12}{4} = \boxed{3}$$

Exercise

CONSIDER THE FOLLOWING DATASET

t	1	2	3	4
u(t)	-1	-1	-1	-1
y(t)	2	-2	0	-1

AND A MODEL CLASS:

$$\mathcal{M}(\Theta): y(t) = a y(t-1) + b u(t-1) + s(t)$$

$$s(t) \sim \mathcal{N}(0, \lambda^2)$$

⇒ FIND $\Theta^* = [a^* \ b^*]$ WHICH MINIMIZES $J_N(\Theta) = \frac{1}{N} \sum (y(t) - \hat{y}(t|t-1))^2$

STEPS:

1. COMPUTE PREDICTOR
2. COMPUTE PREDICTED SAMPLES
3. COMPUTE $J_N(a, b)$
4. MINIMIZE $J_N(a, b)$

1. PREDICTOR

$$y(t) = \frac{\boxed{b}}{\boxed{1 - a z^{-1}}} \cdot u(t-1) + \frac{\boxed{1}}{\boxed{1 - a z^{-1}}} \cdot s(t)$$

$\begin{matrix} B(z) & & C(z) \\ \swarrow & & \swarrow \\ b & & 1 \\ \downarrow & & \downarrow \\ \boxed{1 - a z^{-1}} & & \boxed{1 - a z^{-1}} \\ \uparrow & & \uparrow \\ A(z) & & A(z) \end{matrix}$

SO WE CAN WRITE THE PREDICTOR: $\frac{1}{1} \cdot 1\text{-STEP} \sim \frac{C(z) - A(z)}{C(z)} \cdot 1\text{-STEP ANSWER!}$

$$\hat{y}(t+1|t) = \frac{B(z) \cdot E(z)}{C(z)} \cdot u(t-1+1) + \frac{R(z)}{C(z)} \cdot y(t+1)$$

$$\hat{y}(t+1|t) = b \cdot u(t) + a \cdot y(t)$$

NOTE: THIS APPLIED FORMULA FOR THE PREDICTOR HOLDS ONLY IF:

$\frac{C(z)}{A(z)}$ IS WRITTEN IN CANONICAL FORM!

- SAME ORDER ✓
- MONIC ✓
- COPRIME ✓
- ROOTS WISE ✓

? → WE HAVE TO CHECK IT AFTER WE FIND THE PARAMETER "a"

2. COMPUTE PREDICTIONS

$$y(2|1) = a \cdot y(1) + b \cdot u(1) = 2a - b$$

$$y(3|2) = a \cdot y(2) + b \cdot u(2) = -2a - b$$

$$y(4|3) = a \cdot y(3) + b \cdot u(3) = 0 - b$$

3. COMPUTE COST FUNCTION J_N

$$J_N = \frac{1}{N} \sum_{t=2}^4 (y(t) - \hat{y}(t|t-1))^2$$

$$= \frac{1}{3} \left[(-2 - 2a + b)^2 + (0 + 2a + b)^2 + (-1 + b)^2 \right]$$

$$= \frac{1}{3} \left[5 + 8a^2 + 3b^2 + 8a - 6b \right] = J_N(a, b)$$

4. MINIMIZATION OF J_N

$$\frac{\partial J}{\partial a} = 0 \Rightarrow \frac{1}{3} [16a + 8] = 0 \Rightarrow a = -\frac{1}{2}$$

$$\frac{\partial J}{\partial b} = 0 \Rightarrow \frac{1}{3} [6b - 6] = 0 \Rightarrow b = 1$$

~> THE VALUE OF "a" IS CONSISTENT WITH THE ASSUMPTION OF C/A BEING WRITTEN IN CANONICAL FORM

EXERCISE

CONSIDER A PROCESS DESCRIBED BY THE FOLLOWING DATA:

$$\gamma_y(0) = 3 \quad \gamma_y(1) = 1 \quad \gamma_y(k \geq 2) = 0$$

⇒ FIND THE PARAMETERS OF THE FOLLOWING MODEL USING THE PSM

$$y(t) = \theta_1 \cdot y(t-1) + \theta_2 \cdot y(t-2) + \varepsilon(t) \quad \varepsilon(t) \sim \text{WN}(0, \lambda^2)$$

↳ AR(2) model. NOT A CLEAR MODEL CHOICE ...!

IDEA: SIMILAR TO THE PREVIOUS ONE, BUT WE HAVE TO EXPRESS J AS A FUNCTION OF γ_y .

1. PREDICTOR

$$y(t) = \frac{\boxed{1}^{C(z)}}{\boxed{1 - \theta_1 z^{-1} - \theta_2 z^{-2}}^{A(z)}} \cdot \varepsilon(t)$$

$$\downarrow$$
$$\hat{y}(t+1|t) = [\theta_1 + \theta_2 z^{-1}] \cdot y(t) \quad \sim C(z) - A(z)$$

↓

$$\hat{y}(t+1|t) = \theta_1 \cdot y(t) + \theta_2 \cdot y(t-1)$$

2. WRITING J AS FUNCTION OF γ AND θ

$$J = \mathbb{E}[\varepsilon^2(t+1|t)] = \mathbb{E}[(y(t+1) - \hat{y}(t+1|t))^2]$$

|

$$= \mathbb{E}[(y(t+1) - \theta_1 y(t) - \theta_2 y(t-1))^2]$$

|

$$= \mathbb{E}[y^2(t+1)] + \theta_1^2 \mathbb{E}[y^2(t)] + \theta_2^2 \mathbb{E}[y^2(t-1)] +$$

$$- 2\theta_1 \mathbb{E}[y(t+1) \cdot y(t)] - 2\theta_2 \mathbb{E}[y(t+1) \cdot y(t-1)] + 2\theta_1 \theta_2 \mathbb{E}[y(t) \cdot y(t-1)]$$

$$= \gamma_y(0) + \theta_1^2 \cdot \gamma_y(0) + \theta_2^2 \cdot \gamma_y(0) - 2\theta_1 \cdot \gamma_y(1) - 2\theta_2 \cdot \gamma_y(2) + 2\theta_1 \theta_2 \cdot \gamma_y(1)$$

|

$$J = [1 + \theta_1^2 + \theta_2^2] \cdot \gamma_y(0) + [2\theta_1 \theta_2 - 2\theta_1] \gamma_y(1) - 2\theta_2 \gamma_y(2)$$

$$= j(\gamma, \theta)$$

3. Minimize J

$$\frac{\partial J}{\partial \theta_1} = 0 \Rightarrow 2\theta_1 \gamma_y(0) + [2\theta_2 - 2] \gamma_y(1) = 0 \rightarrow 6\theta_1 + 2\theta_2 = 2$$

$$\frac{\partial J}{\partial \theta_2} = 0 \Rightarrow 2\theta_2 \gamma_y(0) + 2\theta_1 \cdot \gamma_y(1) - 2\gamma_y(2) = 0 \rightarrow 6\theta_2 + 2\theta_1 = 0$$

LET'S SOLVE THIS SYSTEM:

$$\begin{bmatrix} 6 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

↓

$$\begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \frac{1}{32} \begin{bmatrix} 6 & -2 \\ -2 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 3/8 \\ -1/8 \end{bmatrix} = \theta^*$$

FINAL CHECK ON PROCESS STATIONARITY / CANONICAL FORM:

$$z^2 - 3/8 z + 1/8 = 0 \Rightarrow z_{1,2} = \frac{3}{16} \pm \sqrt{\frac{9}{256} - \frac{1}{8}}$$

$$= \frac{3}{16} \pm j \cdot \frac{\sqrt{23}}{16}$$

ABSOLUTE VALUES: $\sqrt{\frac{9}{256} + \frac{23}{256}} = \sqrt{\frac{32}{256}} < 1 \Rightarrow$ INSIDE UNITARY CIRCLE ✓

HOW DO WE COMPUTE λ^2 ?

$$\lambda^2 = \text{Var} [E(t+1|t)] = J(\theta^*) = \left[1 + \frac{9}{64} + \frac{1}{64} \right] \cdot 3 - \left[2 \cdot \frac{3}{8} \cdot \frac{1}{8} + 2 \cdot \frac{3}{8} \right] \cdot 1$$

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EXERCISE

CONSIDER THE S.S.P GENERATED BY:

$$S: y(t) = e(t) + \frac{1}{2} e(t-1) \quad e \sim \text{WN}(0, 1)$$

AND CONSIDER THE MODEL CLASS.

$$M(\theta): y(t) = \eta(t) + a \eta(t-1) \quad \eta \sim \text{WN}(0, \lambda^2)$$

⇒ COMPUTE THE VALUE θ^* WHICH MINIMIZES THE COST FUNCTION

$$J(\theta) = E[(y(t) - \hat{y}(t|t-1, \theta))^2]$$

1. PREDICTOR

$$y(t) = \frac{1 + a z^{-1}}{1} \cdot \eta(t)$$

C(z)
A(z)

$$\hat{y}(t|t-1) = \frac{a z^{-1}}{1 + a z^{-1}} \cdot y(t) = \frac{a}{1 + a z^{-1}} \cdot y(t-1)$$

↓ RECURSIVE EQUATION

$$\hat{y}(t+1|t) = -a \hat{y}(t-1|t-2) + a \cdot y(t-1)$$

IDEA: THE COST FUNCTION IS THE VARIANCE OF $E(t|t-1)$. THE IDEA IS TO EXPRESS $E(t)$ AS A S.P. FOR $y(t)$ AND COMPUTE ITS VARIANCE AS WE DID FOR NORMAL SSP.

2. COST FUNCTION

$$J = E[E(t+1|t)^2] = \text{Var}[E(t+1|t)]$$

$$E(t) = y(t) - \hat{y}(t) = y(t) + a \hat{y}(t-1) - a \cdot y(t-1)$$

$$= y(t) - a [y(t-1) - \hat{y}(t-1)]$$

$$E(t) = y(t) - a \cdot E(t-1)$$

$$E(t) = e(t) + \frac{1}{2} e(t-1) - a E(t-1)$$

ARMA PROCESS!

ALTERNATIVE WAY, BASED ON T.F.:

$$E(t) = y(t) - \hat{y}(t) \quad \text{D.F. OF } \hat{y} \text{ IN T.F.}$$

$$= \left[y(t) - \frac{a \cdot z^{-1}}{1 + a z^{-1}} \cdot y(t) \right] = \left[1 - \frac{a z^{-1}}{1 + a z^{-1}} \right] \cdot y(t) \quad \text{D.F. OF } y(t) \text{ IN T.F.}$$

$$= \frac{1}{1 + a z^{-1}} \cdot [1 + \frac{1}{2} z^{-1}] e(t)$$

↓

$$\mathcal{E}(t) = \frac{1 + 1/2 z^{-1}}{1 + a z^{-1}} \cdot e(t)$$

SAME ARMA PROCESS!

THE COST FUNCTION IS THE VARIANCE OF A ARMA SSP:

$$\begin{aligned} J(\theta) &= \mathbb{E}[\mathcal{E}^2(t)] = \mathbb{E}\left[(-a \mathcal{E}(t-1) + e(t) + 1/2 e(t-1))^2\right] \\ &= a^2 \mathbb{E}[\mathcal{E}^2(t-1)] + \mathbb{E}[e^2(t)] + 1/4 \mathbb{E}[e^2(t-1)] + \\ &\quad -2a \mathbb{E}[\mathcal{E}(t-1) \cdot e(t)] - a \cdot \mathbb{E}[\mathcal{E}(t-1) \cdot e(t-1)] + \mathbb{E}[e(t) \cdot e(t-1)] \\ \gamma_{\mathcal{E}}(0) &= a^2 \cdot \gamma_{\mathcal{E}}(0) + \lambda_e^2 + 1/4 \cdot \lambda_e^2 + 0 - a \lambda_e^2 + 0 \end{aligned}$$

$$\mathbb{E}[\mathcal{E}(t-1) \cdot e(t)] = 0 \quad \text{BECAUSE } \mathcal{E}(t-1) = f(e(t-1), e(t-2), \dots)$$

$$\mathbb{E}[\mathcal{E}(t-1) \cdot e(t-1)] = \mathbb{E}\left[(-a \mathcal{E}(t-2) + e(t-1) + 1/2 e(t-2)) \cdot e(t-1)\right] = \lambda_e^2$$

SO THE COST FUNCTION $J = \text{Var}[\mathcal{E}(t)] = \gamma_{\mathcal{E}}(0)$ IS EQUAL TO:

$$\gamma_{\mathcal{E}}(0) = \frac{5/4 - a}{1 - a^2} = J(\theta)$$

3. MINIMIZATION OF $J(\theta)$

$$\frac{\partial J}{\partial a} = \frac{-1}{1 - a^2} + \frac{(5/4 - a) \cdot 2a}{(1 - a^2)^2} = \frac{-a^2 + 5/2 a - 1}{(1 - a^2)^2}$$

THIS MODEL WOULDN'T BE IN CANONICAL FORM!

$$\rightarrow \frac{\partial J}{\partial a} = 0 \quad a^2 - 5/2 a + 1 = 0 \rightarrow a_{1,2} = \begin{cases} 8/4 = 2 & \times \\ 1/2 & \checkmark \end{cases}$$

$$a = 1/2$$

AS EXPECTED! MODEL CLASS COINCIDES WITH THE SYSTEM CLASS

LET'S COMPUTE THE NOISE VARIANCES:

$$\text{Var}[v(t)] = J(a^*) = \frac{5/4 - 1/2}{1 - 1/4} = \frac{3/4}{3/4} = 1 = \lambda^2$$